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A Hands-On Introduction to Single-Cell RNA-Seq Analysis

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McGill | CDSI Computational and Data Systems Institute

D2R DNA TO RNA | HeDS

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What You Will Learn Today

Understand why single-cell analysis is powerful and how it solves problems that bulk RNA-seq misses.

Familiarize yourself with the complete computational workflow for a single-cell RNA-seq experiment.

Gain an understanding of key concepts like Normalization, Dimensionality Reduction, and Doublet Detection.

Learn how to use marker genes to identify and annotate cell clusters.

What You Will NOT Learn Today

Processing raw sequencing files (FASTQs) to create the count matrix.

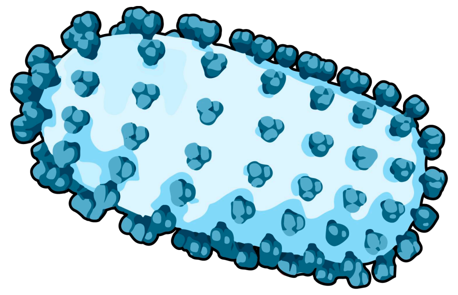
We will not cover complex methods for batch correction or integrating multiple different experiments.

Statistically valid differential expression analysis (like DESeq2) which requires true biological replicates.

Other single-cell data types like ATAC-seq (chromatin), CITE-seq (protein), or spatial transcriptomics.

We will not get into more complex topics like trajectory inference (pseudotime), RNA velocity, or deep algorithm theory.

The Mystery of Two Patients: Why Do Immune Systems Respond So Differently?



Two patients are exposed to the same virus. Both are the same age, same health status, no pre-existing conditions.

David: Mild symptoms, recovers in 5 days.

Emily: Severe disease, hospitalized, requires intensive care.

The Mystery of Two Patients: Why Do Immune Systems Respond So Differently?

What's different?

Both have immune cells fighting the infection. Both show "immune activation" in standard blood tests

The Question:

Why Did Emily Get Sicker?

The Mystery of Two Patients: Why Do Immune Systems Respond So Differently?

What's different?

Both have immune cells fighting the infection. Both show "immune activation" in standard blood tests

Hypothesis

1: Emily has fewer protective immune cells (like T cells)

2: Emily has different cell types or cell states that David doesn't have

Hypothesis 1 **Do They Have Different Amounts of Immune Cells?**

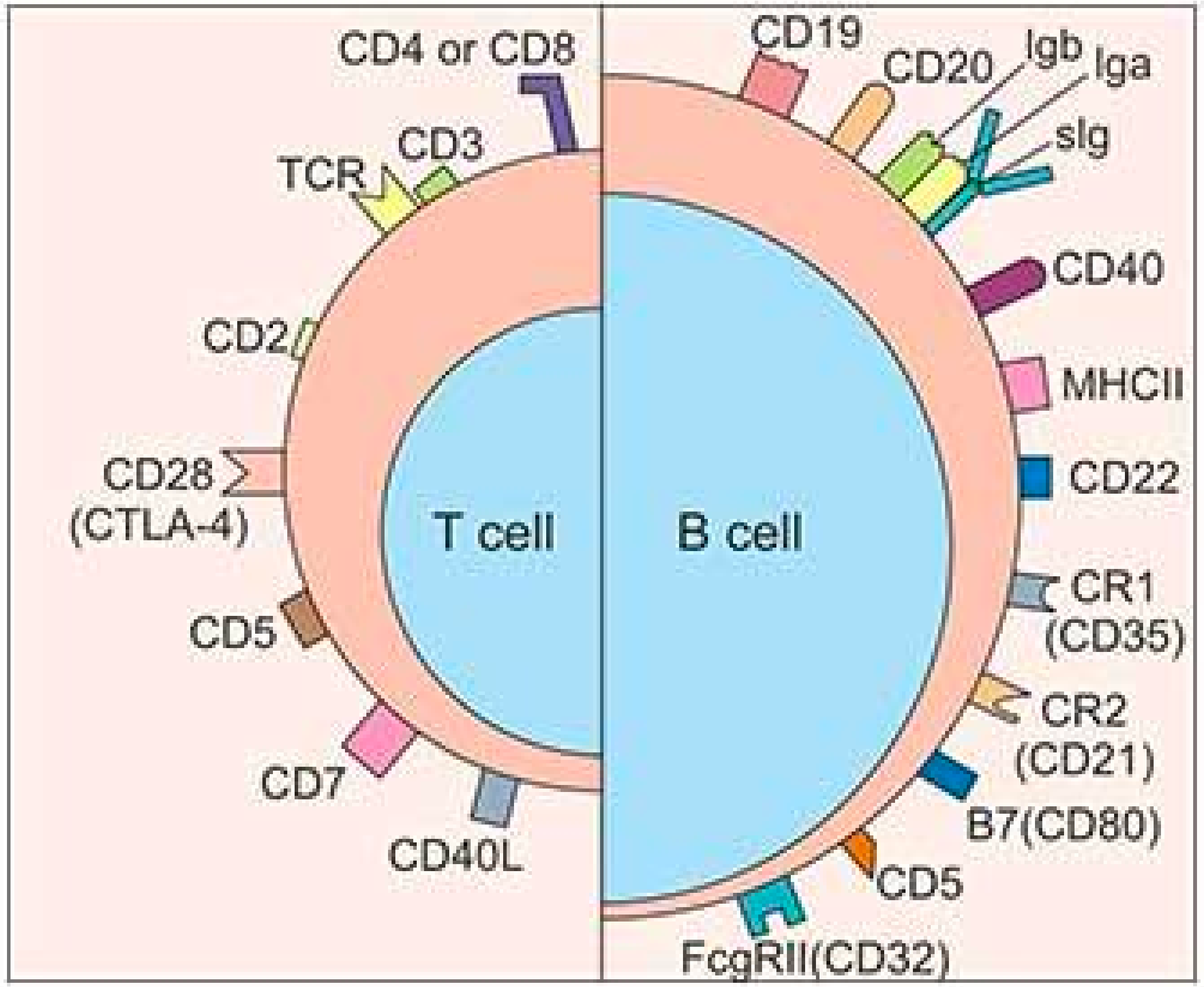
We can look at:

T cell marker genes (CD3D, CD8A)

B cell marker genes (CD19, MS4A1)

Monocyte marker genes (CD14, CD68)

...



https://www.cusabio.com/Cell-Marker/Cell-Surface-Marker.html?srsltid=AfmBOooBpppX0Tetlvfpum-aS3zQLsMEqM-GTkUo0J9lcPLWO2Njjxq6

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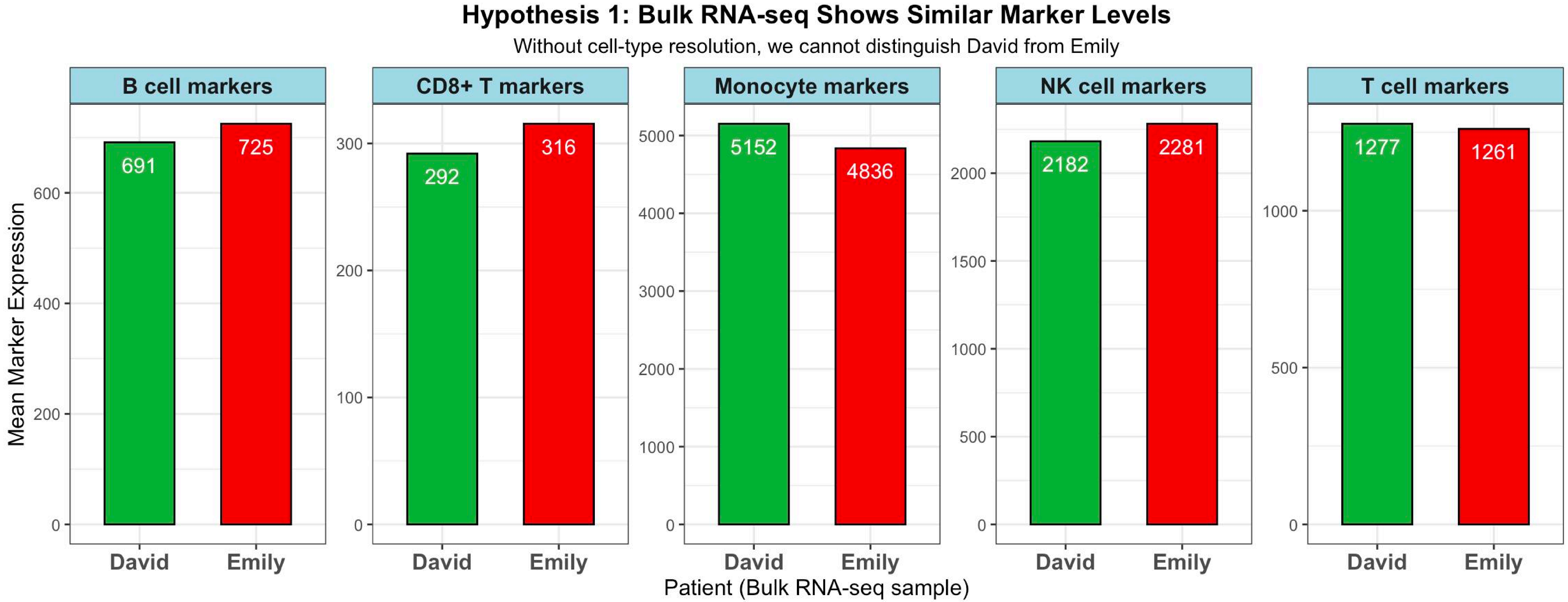

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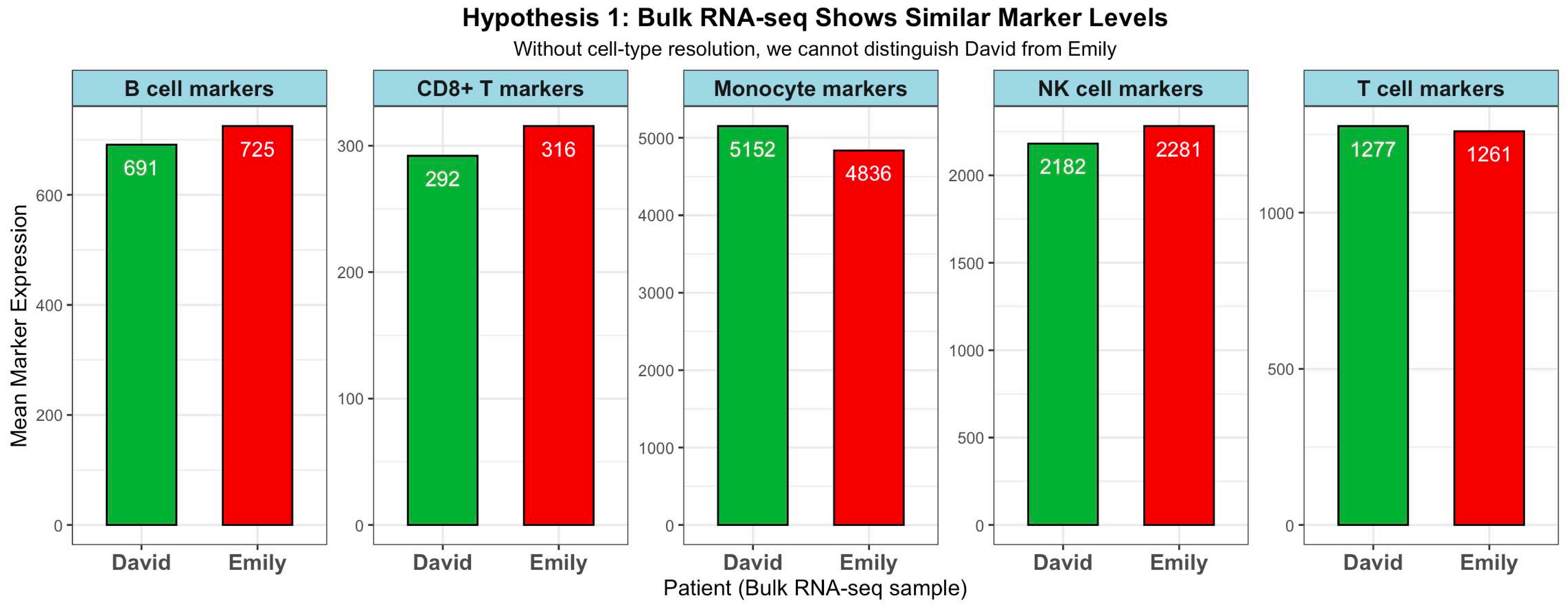
Hypothesis 1 **Do They Have Different Amounts of Immune Cells?**



Both patients have similar levels of immune cell marker expression level.

Stop and think

What factor we did not consider here?

Stop and think

What factor we did not consider here?

This doesn't tell us the actual NUMBERS of each cell type. It only tells us average gene expression.

Hypothesis 2 **Emily has different cell types or cell states that David doesn't have**

Bulk sequencing averages everything together into one number per gene.

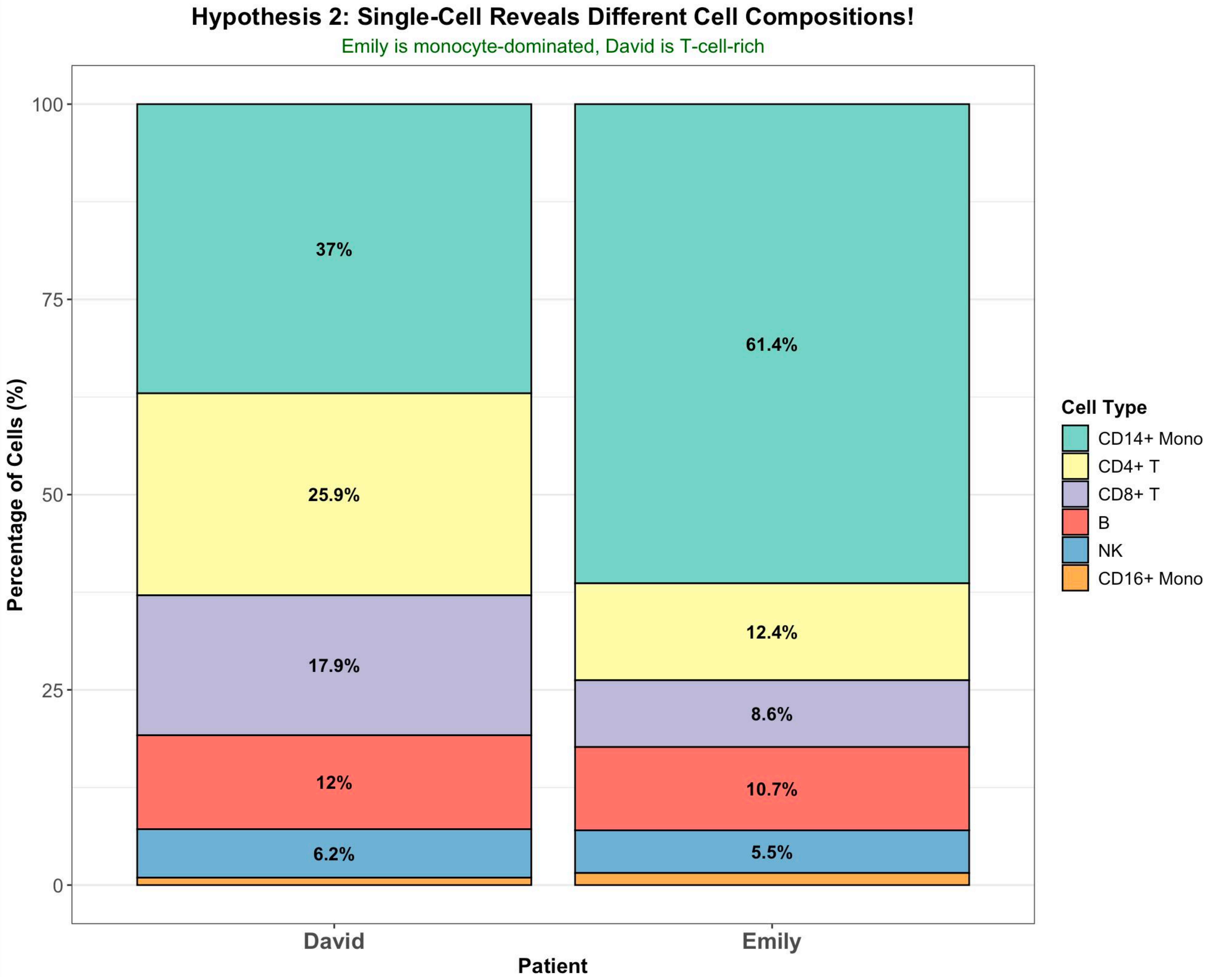
If Emily has a rare population of harmful cells (say, 5% of her blood), their signal gets averaged away with the other 95%.

We Need a Different Approach:

Instead of averaging all cells together...

We need to measure each cell individually.

Hypothesis 2 **Emily has different cell types or cell states that David doesn't have**



The New Approach:

Instead of grinding all blood cells together, we measure each cell separately.

Each cell gets its own gene expression profile.

What This Allows Us To Do:

- Count exactly how many of each cell type are present
- See what each cell type is actually doing (which genes they express)
- Discover rare cell populations (even if they're only 1-5% of the sample)
- Compare cell-by-cell between two patients

The New Approach:

To analyze cells individually, we need to label their RNA so we know which cell it came from.

Early methods (2009-2013): Manual picking - literally isolate cells one by one into tubes (10-100 cells)

Plate-based methods (Smart-seq, SMART-seq2): Automated but still limited (Hundreds of cells)

Droplet-based methods (2015-present): The game changer!

- Drop-seq, inDrop, 10x Genomics Chromium
- Thousands to tens of thousands of cells per run
- Much cheaper per cell

The New Approach:

Today's Standard: 10x Genomics Chromium

High throughput (1,000-10,000 cells per sample)

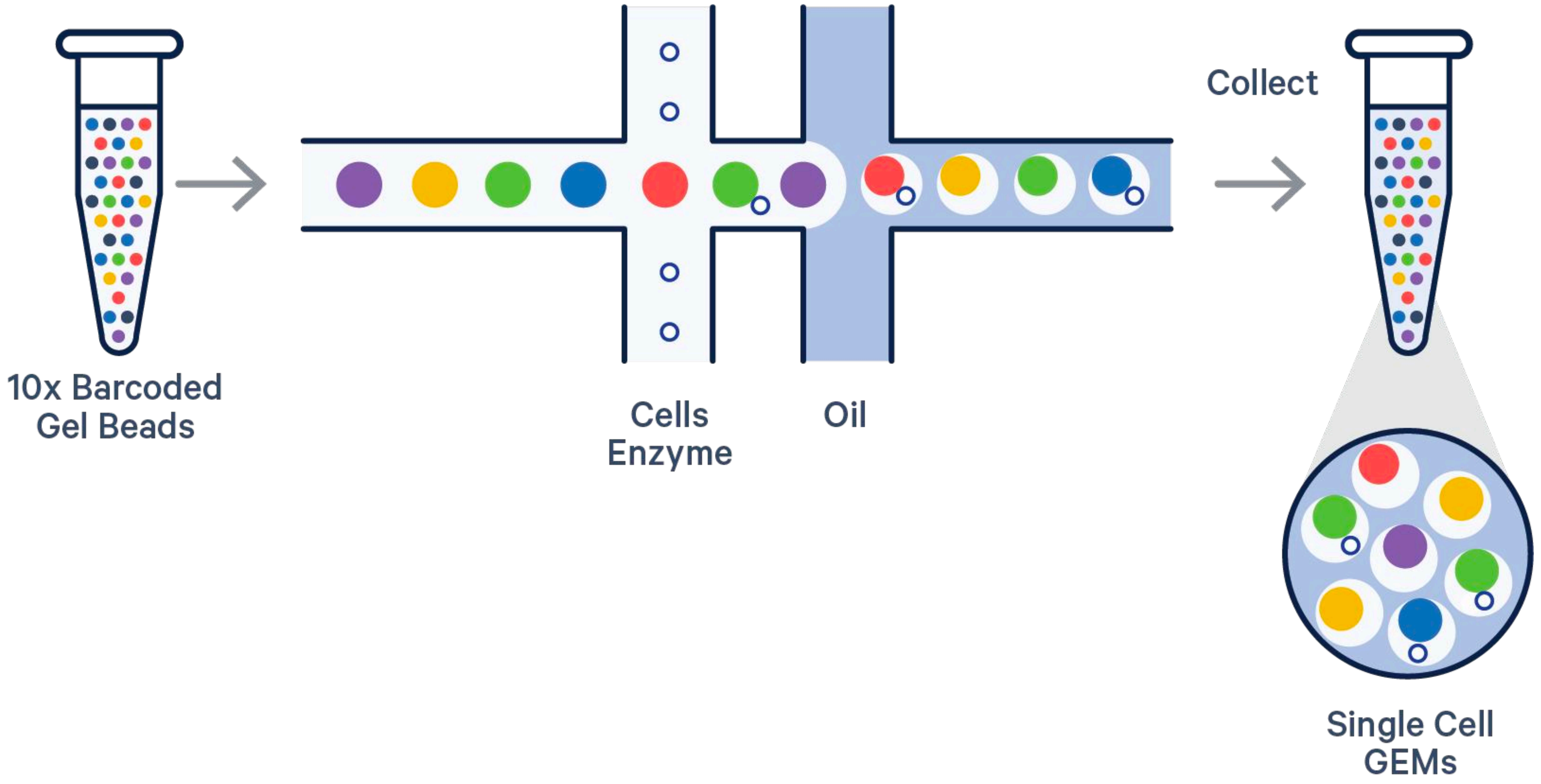
Relatively low cost (~$1-3 per cell)

The Core Principle

Trap single cells in tiny oil droplets with barcoded beads.

Each cell's RNA gets a unique barcode.

Sequence everything together and sort by barcode.



https://www.10xgenomics.com/blog/the-next-generation-of-single-cell-rna-seq-an-introduction-to-gem-x-technology

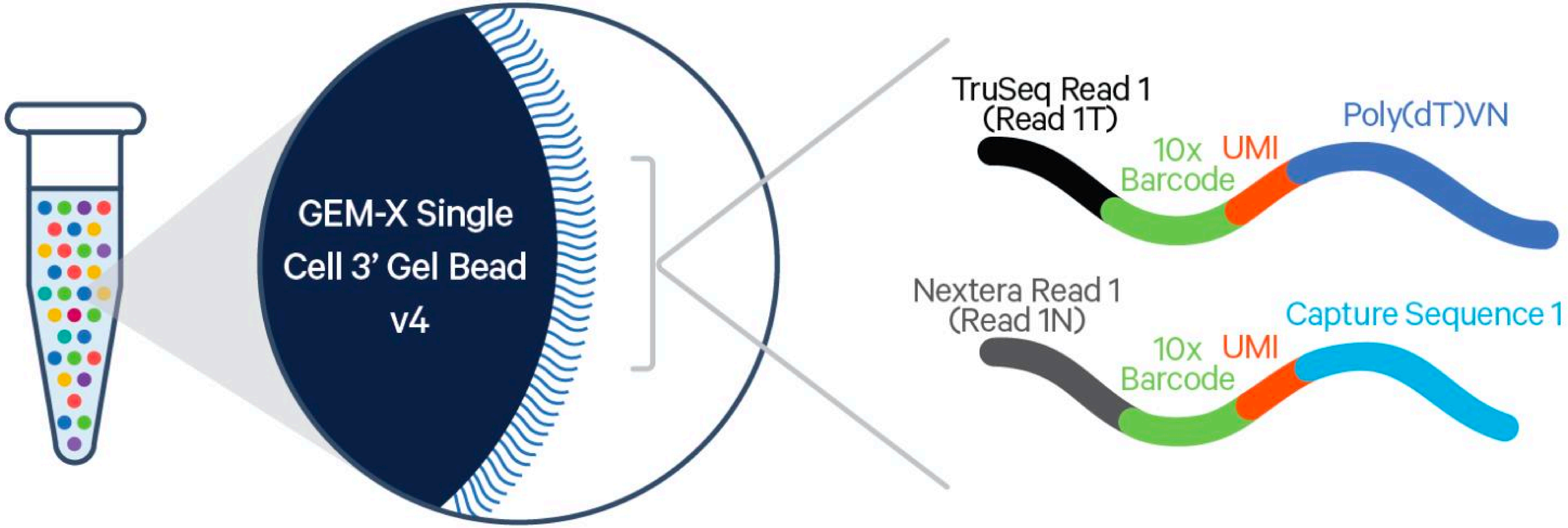


Figure 2. Schematic diagram of a GEM-X Single Cell 3' Gel Bead. Every Gel Bead is coated with oligos containing an Illumina TruSeq Read 1 (read 1 sequencing primer, Read 1T), 16 nt 10x Barcode, 12 nt unique molecular identifier (UMI), and 30 nt poly(dT)VN.

The mixture is composed of a cell suspension with lysis buffer, Gel Beads coated with oligonucleotides—which include a 10x Barcode that marks each RNA molecule's cell of origin and a unique molecular identifier (UMI)

Today's Standard: 10x Genomics Chromium

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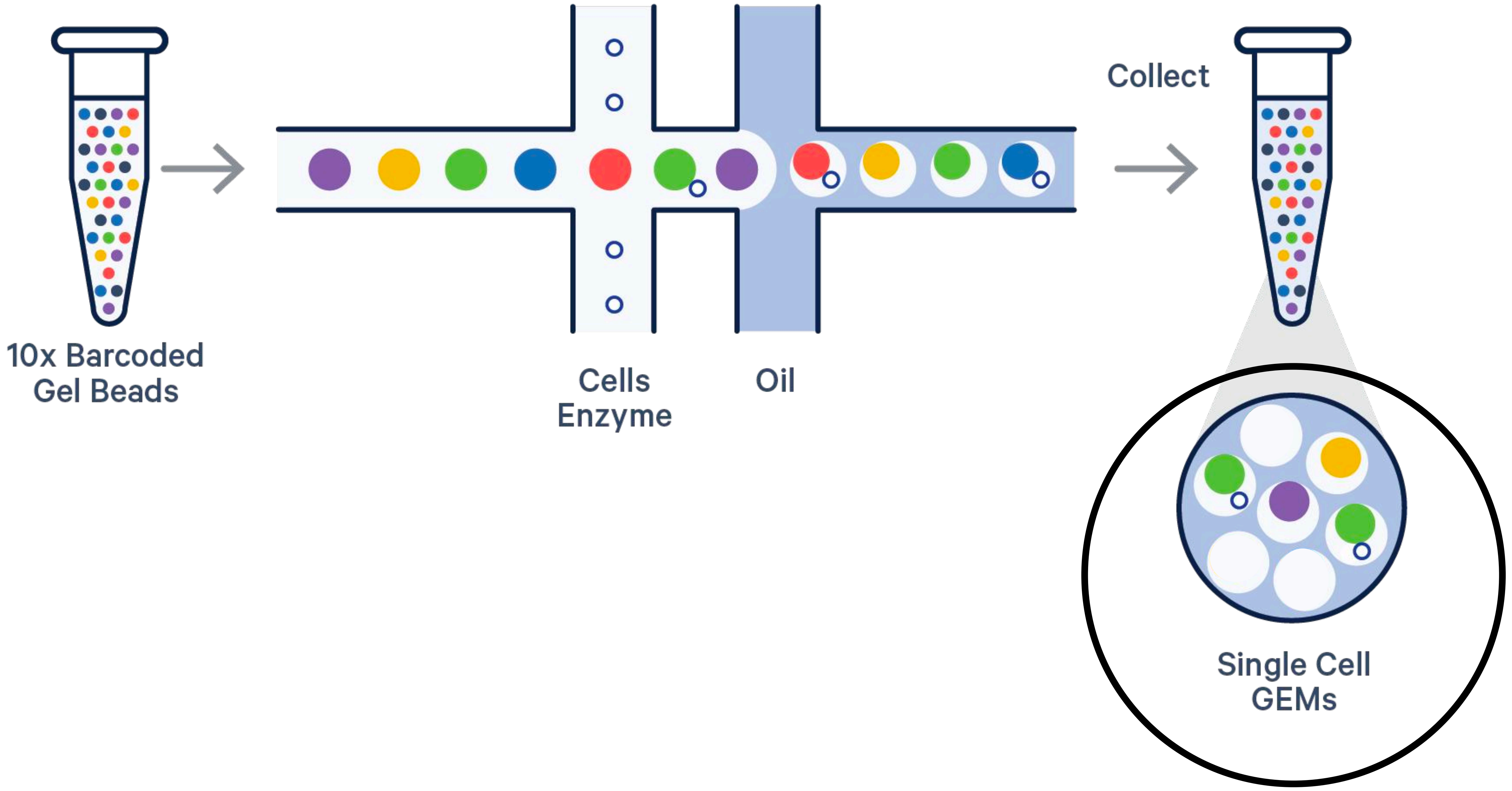

https://www.10xgenomics.com/blog/the-next-generation-of-single-cell-rna-seq-an-introduction-to-gem-x-technology

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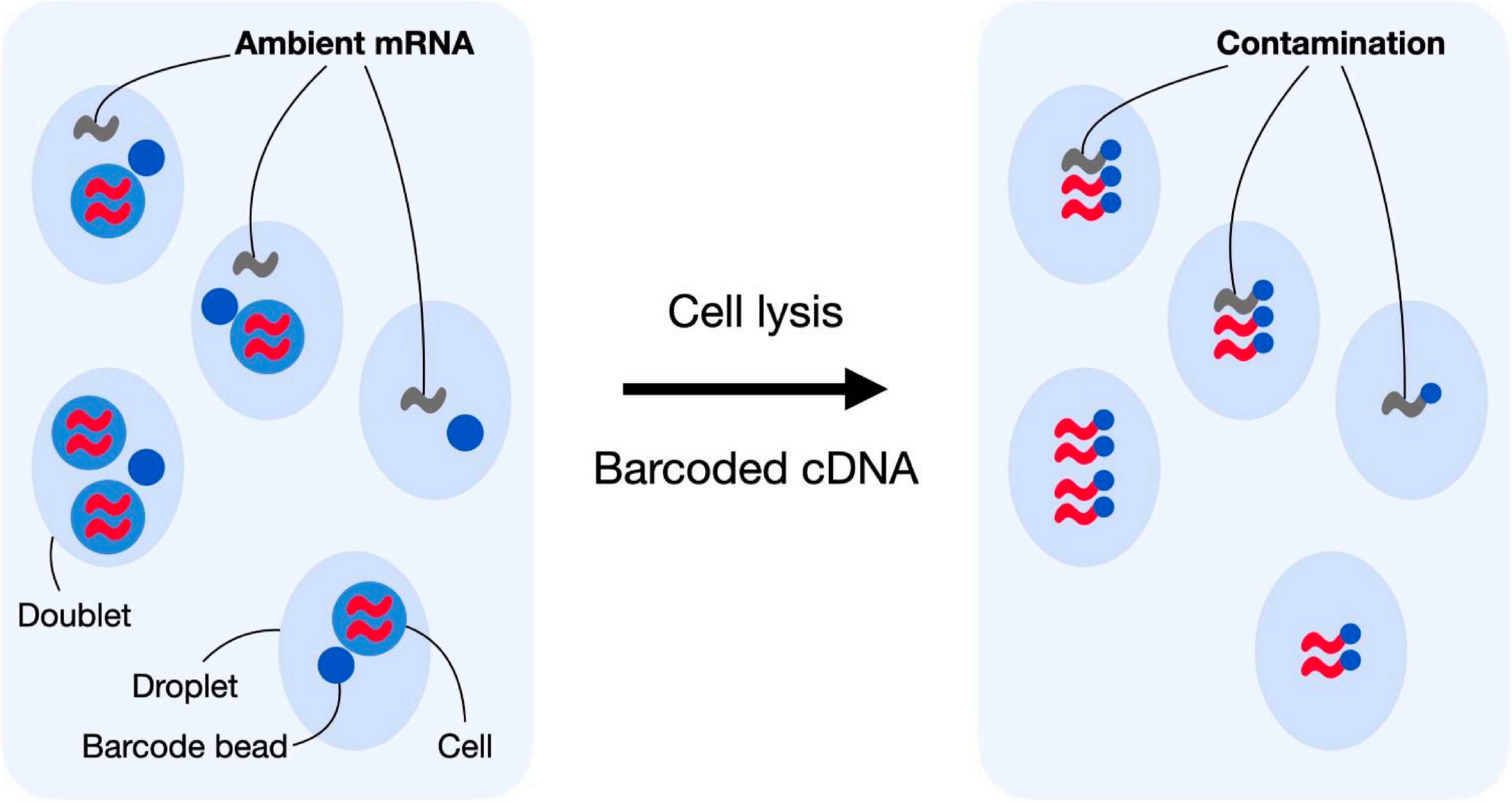
Today's Goal:
Learn the computational workflow to analyze single-cell RNA-seq data.
Understand the strategic decisions at each step.
Critically evaluate assumptions and limitations.

Our Dataset:
Blood samples (PBMCs) from healthy donors (Already loaded in the SeuratData package).
~3,000 cells already processed and sequenced.
We've added simulated "David" and "Emily" labels to demonstrate comparisons.

Step 1.1 - Three Quality Control Covariates

Example of bad data

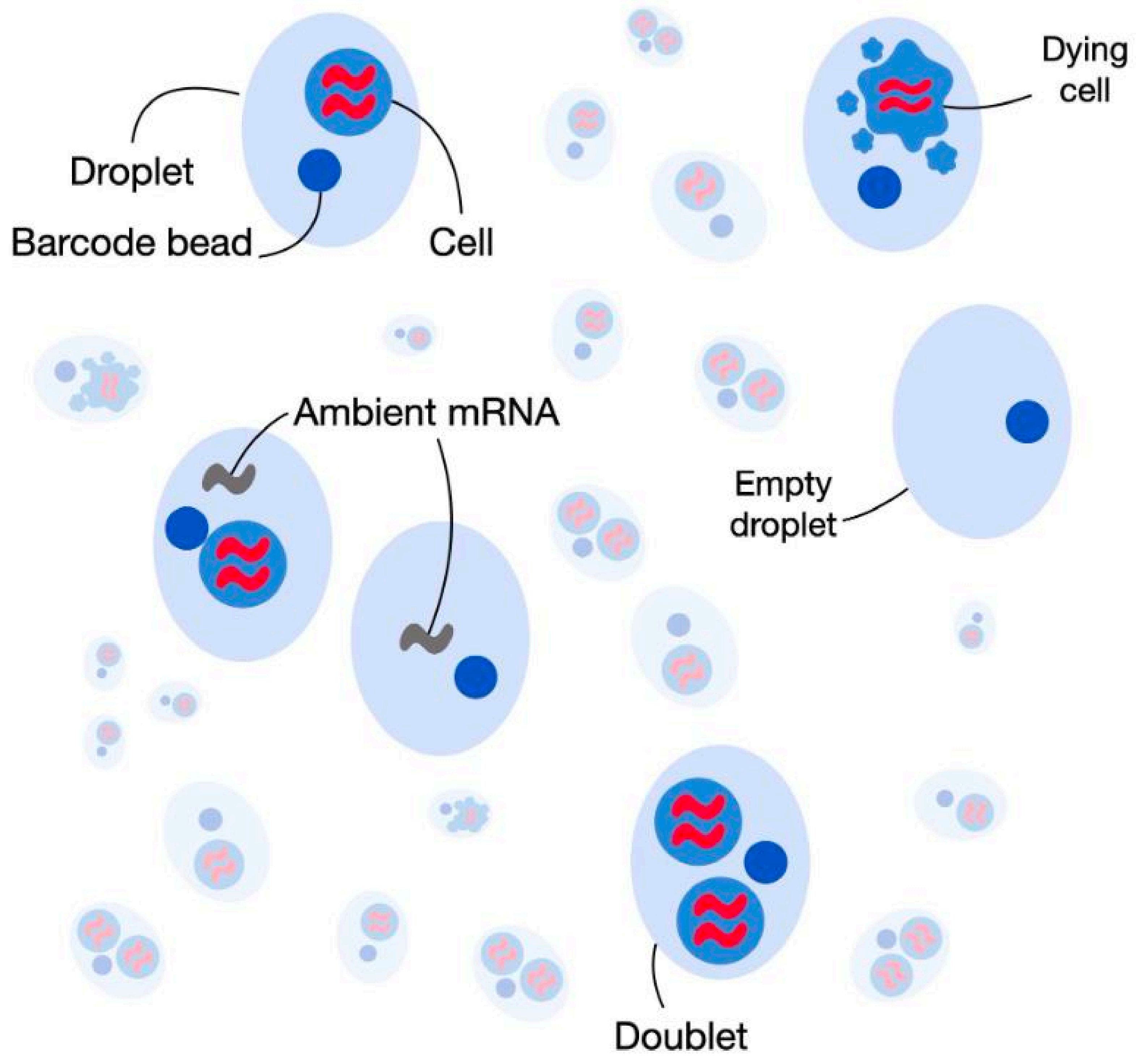


Droplet-based scRNA-seq generates counts of **unique molecular identifiers (UMIs)** for genes across multiple cells and aims to identify the number of molecules for each gene and each cell.

It assumes that each droplet contains mRNA from single cells.

Step 1.1 - Three Quality Control Covariates

Another example of bad data



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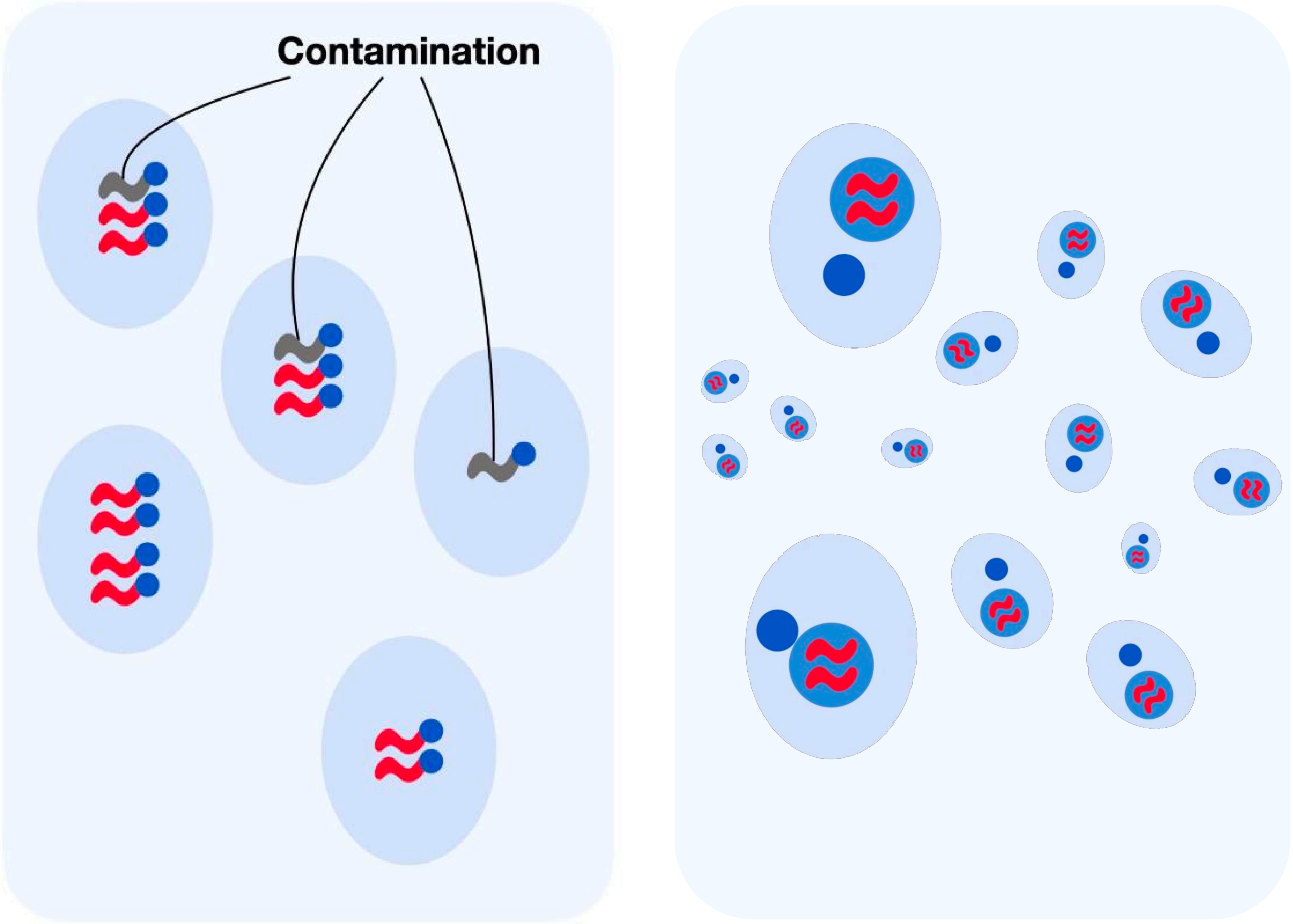
Step 1.1 - Three Quality Control Covariates

<u>Example of bad data</u>   <u>and good data</u>

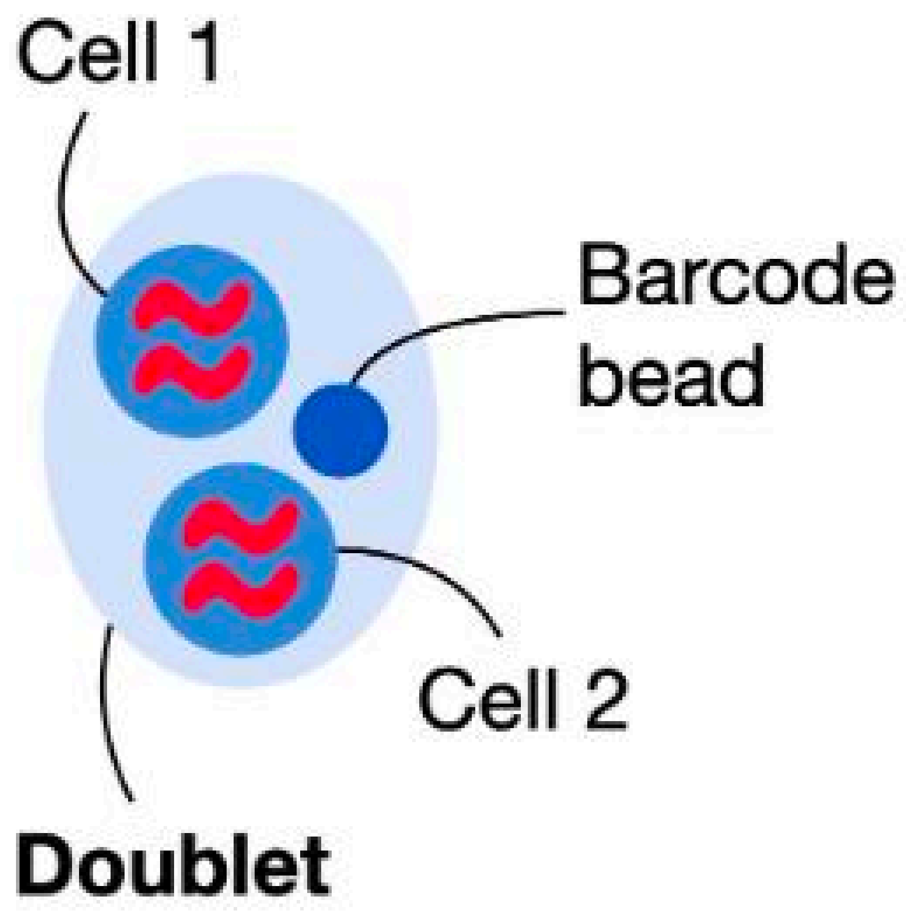


Cell QC is typically performed on the following three QC covariates:

1. The number of counts per barcode (count depth)
2. The number of genes per barcode
3. The fraction of counts from mitochondrial genes per barcode

Step 1.2 - Doublet Detection

Doublets are defined as two cells that are sequenced under the same cellular barcode



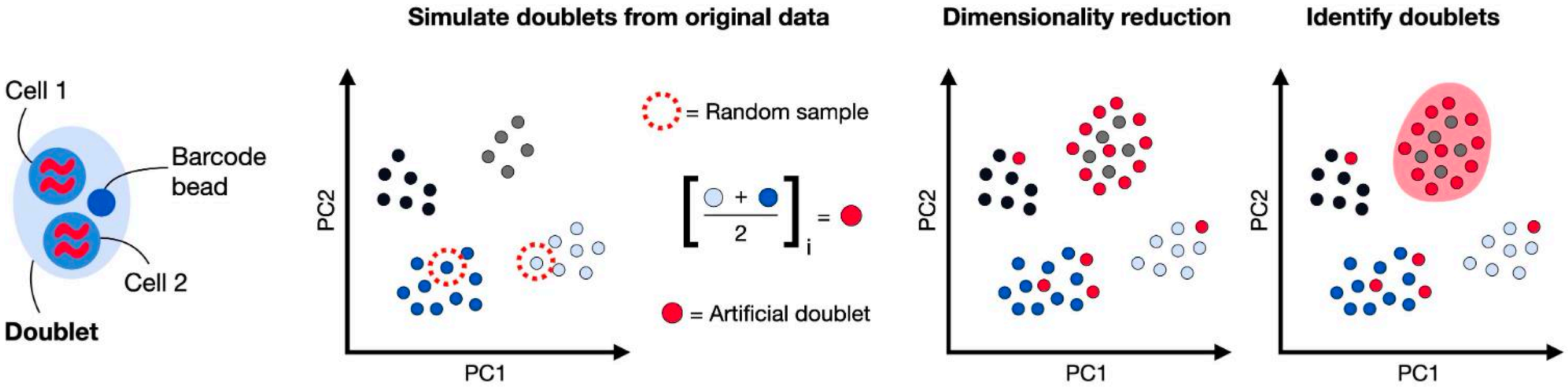
Their identification is crucial as they are most likely misclassified and can lead to distorted downstream analysis steps

Doublets can be either identified through

1. High number of reads and detected features
2. Methods that create artificial doublets and compare these with the existing cell profiles

Step 1.2 - Doublet Detection

Common doublet detection methods like **scDblFinder** generate artificial doublets by randomly subsampling pairs of cells and averaging their gene expression profile to obtain doublet counts.



Step 1.3 - Normalization

```
# This takes a bit longer than other cells
pbmc <- SCTransform(pbmc, vars.to.regress = "percent.mt", verbose = FALSE)
```

The Problem:

Different cells got sequenced to different depths.

Some cells: 2,000 total transcripts detected.

Other cells: 20,000 total transcripts detected.

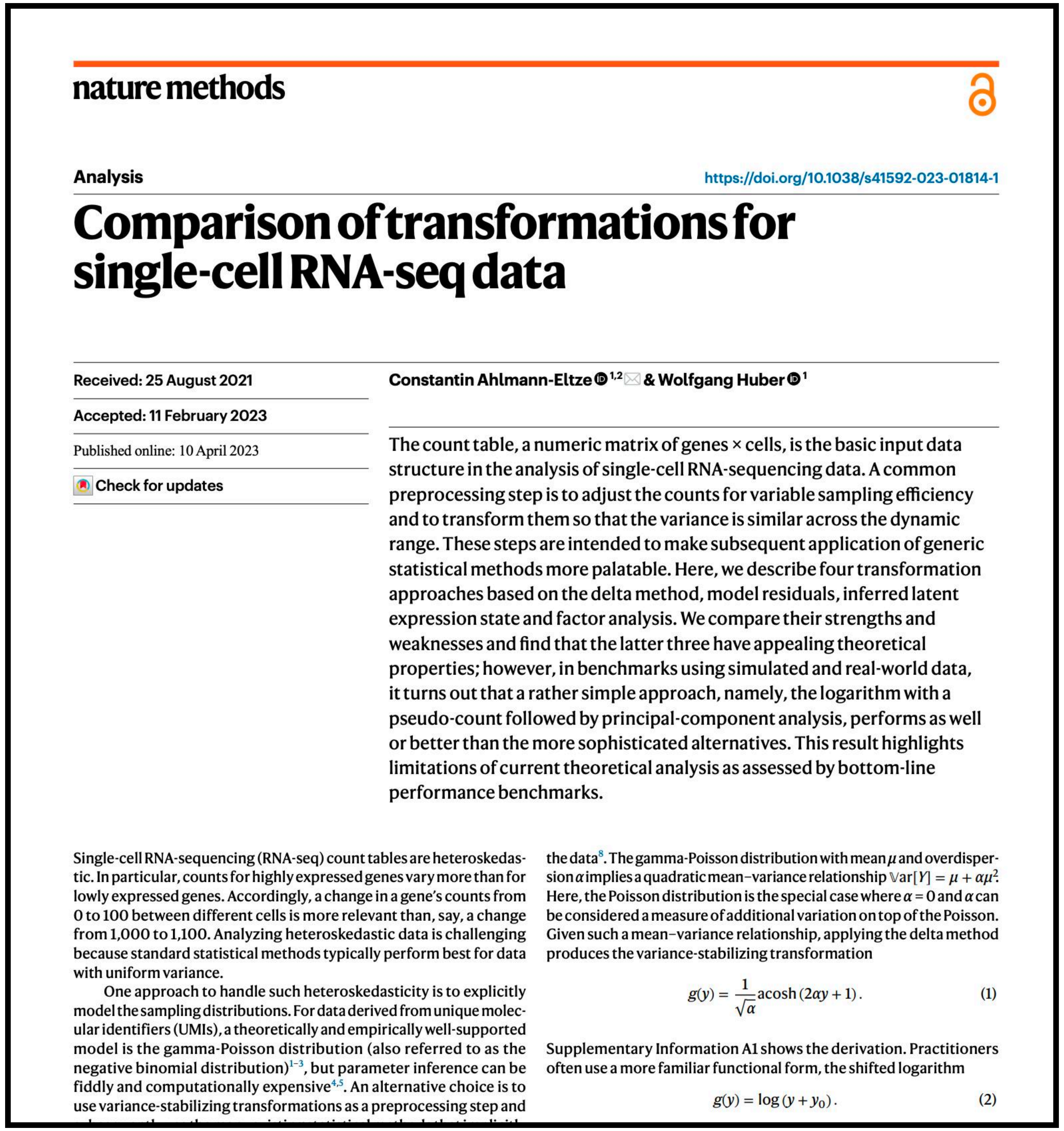
<u>This is just technical randomness, not biology!</u>

Remove technical variation while keeping biological variation and make cells fairly comparable to each other.

Step 1.3 - Normalization

Several normalization techniques are used in practice varying in complexity.

SCTransform is the current best practice for single-cell data, and is a fast normalization technique, outperforms other methods for uncovering the latent structure of the datasets



nature methods

Analysis

https://doi.org/10.1038/s41592-023-01814-1

Comparison of transformations for single-cell RNA-seq data

Received: 25 August 2021

Accepted: 11 February 2023

Published online: 10 April 2023

Check for updates

Constantin Ahlmann-Eltze [1,2] & Wolfgang Huber [1]

The count table, a numeric matrix of genes × cells, is the basic input data structure in the analysis of single-cell RNA-sequencing data. A common preprocessing step is to adjust the counts for variable sampling efficiency and to transform them so that the variance is similar across the dynamic range. These steps are intended to make subsequent application of generic statistical methods more palatable. Here, we describe four transformation approaches based on the delta method, model residuals, inferred latent expression state and factor analysis. We compare their strengths and weaknesses and find that the latter three have appealing theoretical properties; however, in benchmarks using simulated and real-world data, it turns out that a rather simple approach, namely, the logarithm with a pseudo-count followed by principal-component analysis, performs as well or better than the more sophisticated alternatives. This result highlights limitations of current theoretical analysis as assessed by bottom-line performance benchmarks.

Single-cell RNA-sequencing (RNA-seq) count tables are heteroskedastic. In particular, counts for highly expressed genes vary more than for lowly expressed genes. Accordingly, a change in a gene's counts from 0 to 100 between different cells is more relevant than, say, a change from 1,000 to 1,100. Analyzing heteroskedastic data is challenging because standard statistical methods typically perform best for data with uniform variance.

One approach to handle such heteroskedasticity is to explicitly model the sampling distributions. For data derived from unique molecular identifiers (UMIs), a theoretically and empirically well-supported model is the gamma-Poisson distribution (also referred to as the negative binomial distribution)[1–3], but parameter inference can be fiddly and computationally expensive[4,5]. An alternative choice is to use variance-stabilizing transformations as a preprocessing step and

the data[8]. The gamma-Poisson distribution with mean μ and overdispersion α implies a quadratic mean–variance relationship $\mathbb{V}\mathrm{ar}[Y] = \mu + \alpha\mu^2$. Here, the Poisson distribution is the special case where $\alpha = 0$ and α can be considered a measure of additional variation on top of the Poisson. Given such a mean–variance relationship, applying the delta method produces the variance-stabilizing transformation

$$g(y) = \frac{1}{\sqrt{\alpha}} \operatorname{acosh}(2\alpha y + 1). \qquad (1)$$

Supplementary Information A1 shows the derivation. Practitioners often use a more familiar functional form, the shifted logarithm

$$g(y) = \log(y + y_0). \qquad (2)$$

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Step 1.3 - Normalization

What SCTransform Does:

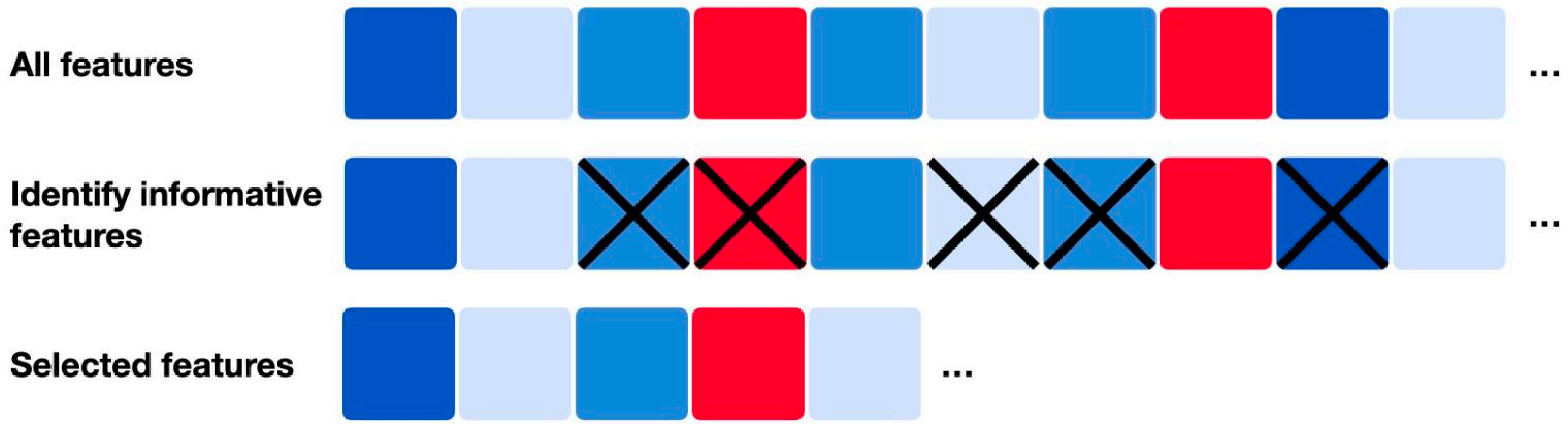
1. Models the relationship between sequencing depth and gene expression.
2. For each gene, calculates expected counts based on cell size.
3. Creates "residuals" - how much each cell deviates from expectation.
4. These residuals become our normalized values.

Step 1.4 - Feature selection

The Problem:

After normalization, we still have ~20,000-30,000 genes.

Many of these genes are NOT informative:



1. Expressed in very few cells
2. Show no variation across cells
3. Mostly contain zeros

Why This Matters:

Including uninformative genes adds noise to downstream analysis.

Slows down computation (PCA, clustering).

Can obscure real biological signals.

Step 1.4 - Feature selection

Stop and think

Why we cannot filter based on Highly Expressed Genes?

Step 1.4 - Feature selection

Select genes with highest average expression.

Simple but misses lowly-expressed but variable genes.

Modern Approach: Deviance-Based Selection

Works directly on raw counts (no normalization needed).

Identifies genes that deviate from constan expression.

Not affected by arbitrary pseudo-count choices.

How Deviance Works:

Calculate deviance: How much does each gene deviate from this null model?

High deviance = Gene varies a lot = Informative!

Low deviance = Gene is constant = Not informative. => house keeping genes!

Step 2 - Dimensionality Reduction

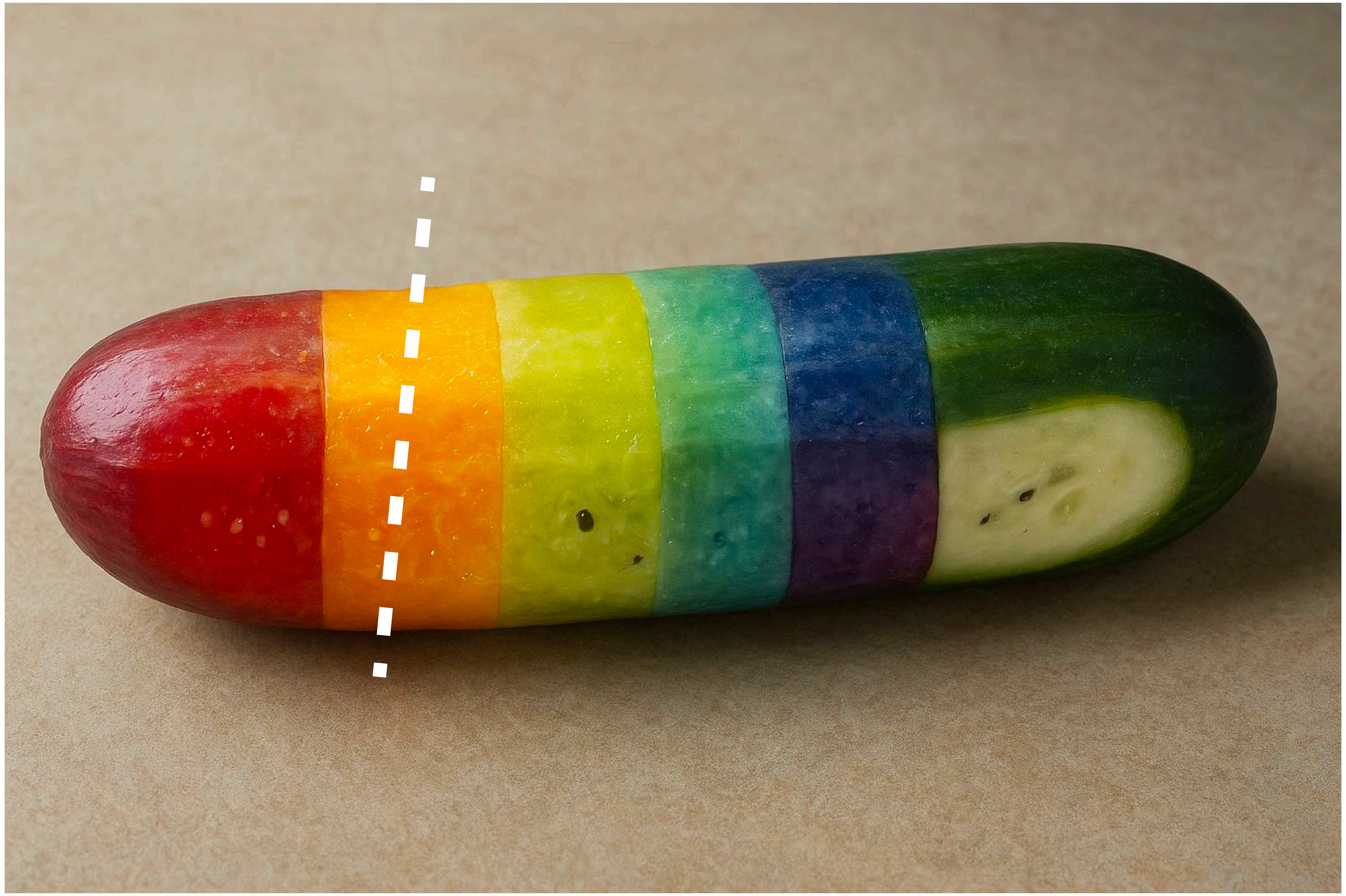
Stop and think

Why do we need to reduce dimensionality?

Step 2 - Dimensionality Reduction

Image generated using OpenAI Chat-GPT5

Step 2 - Dimensionality Reduction



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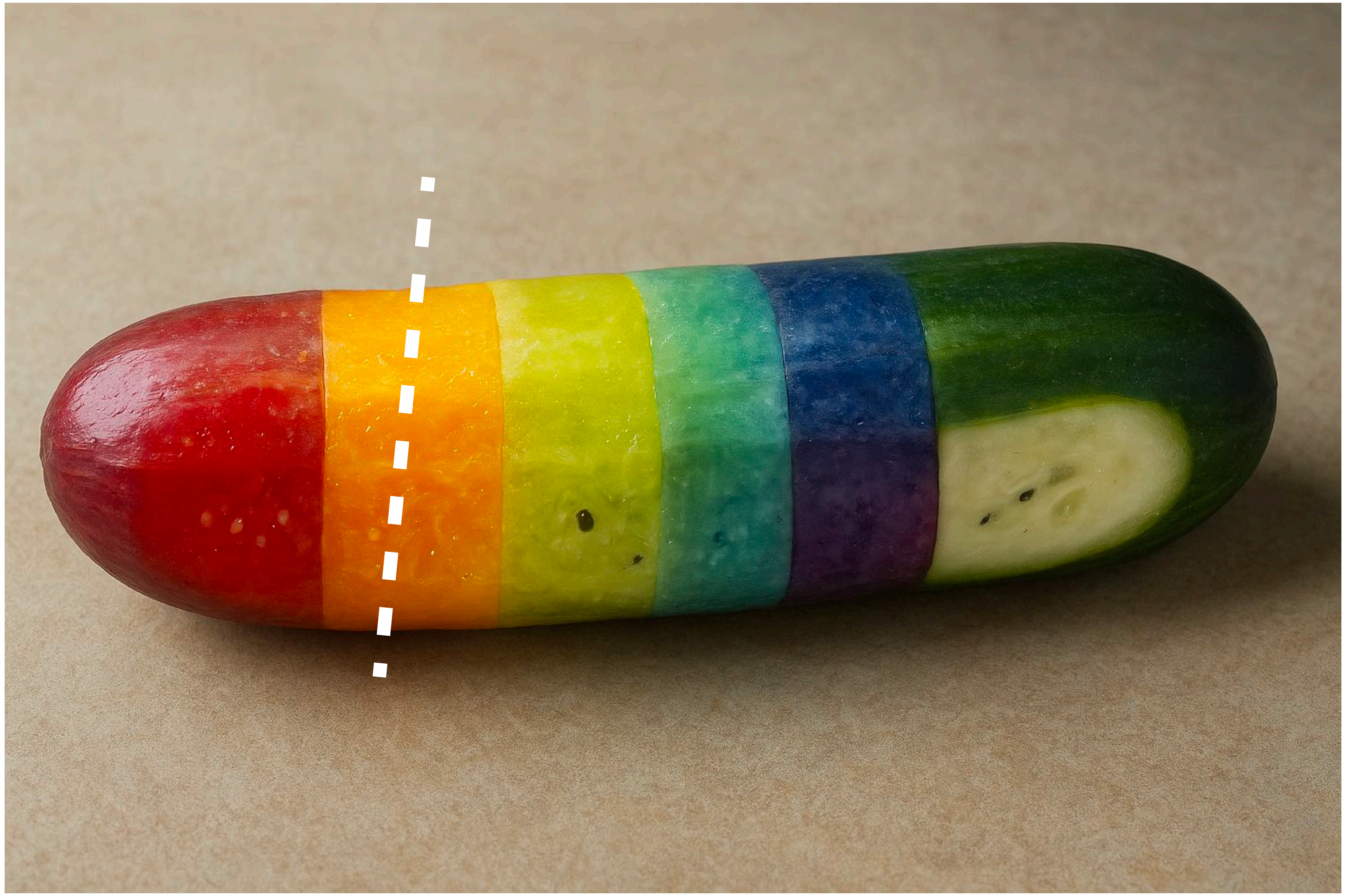

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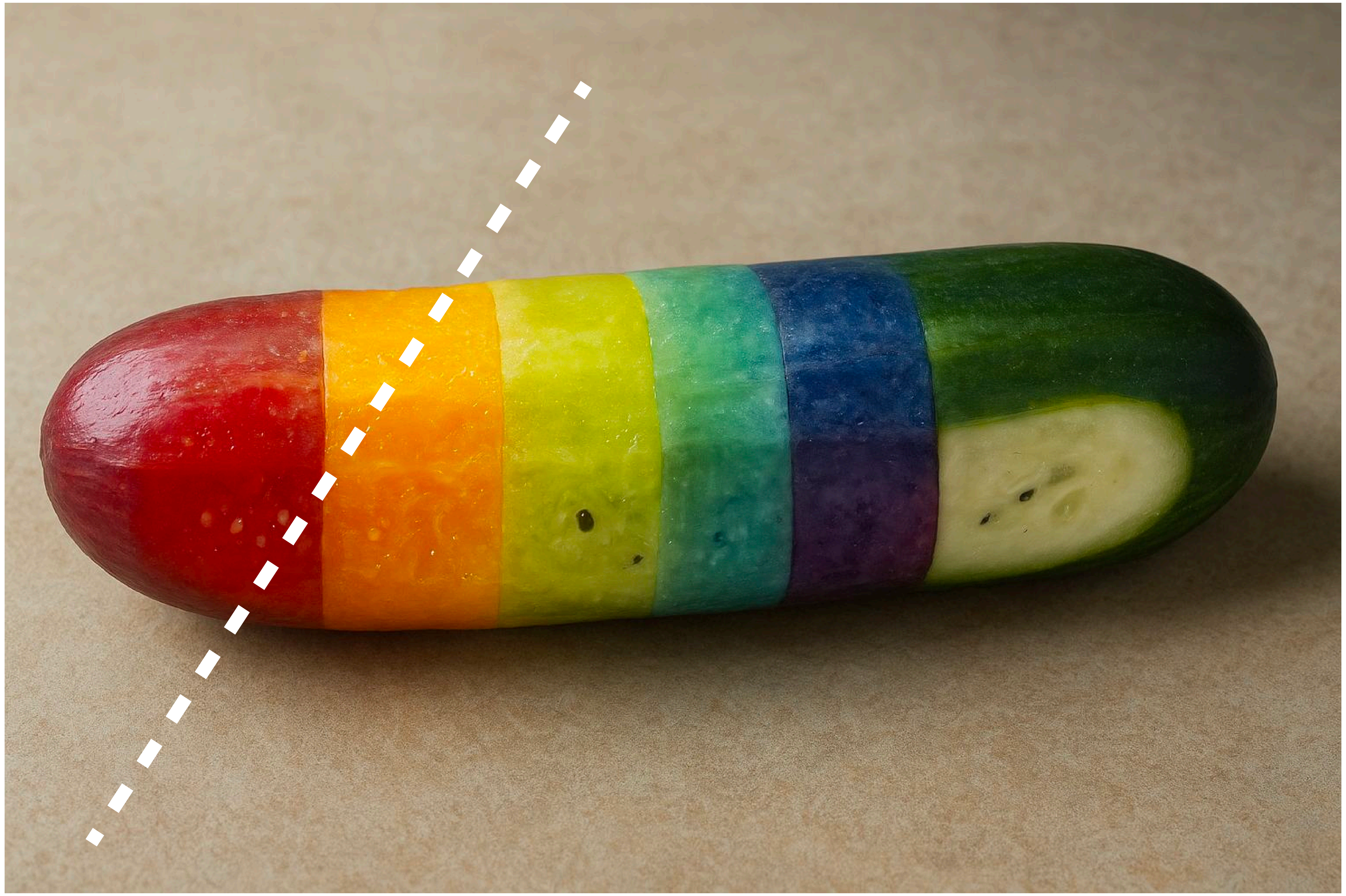

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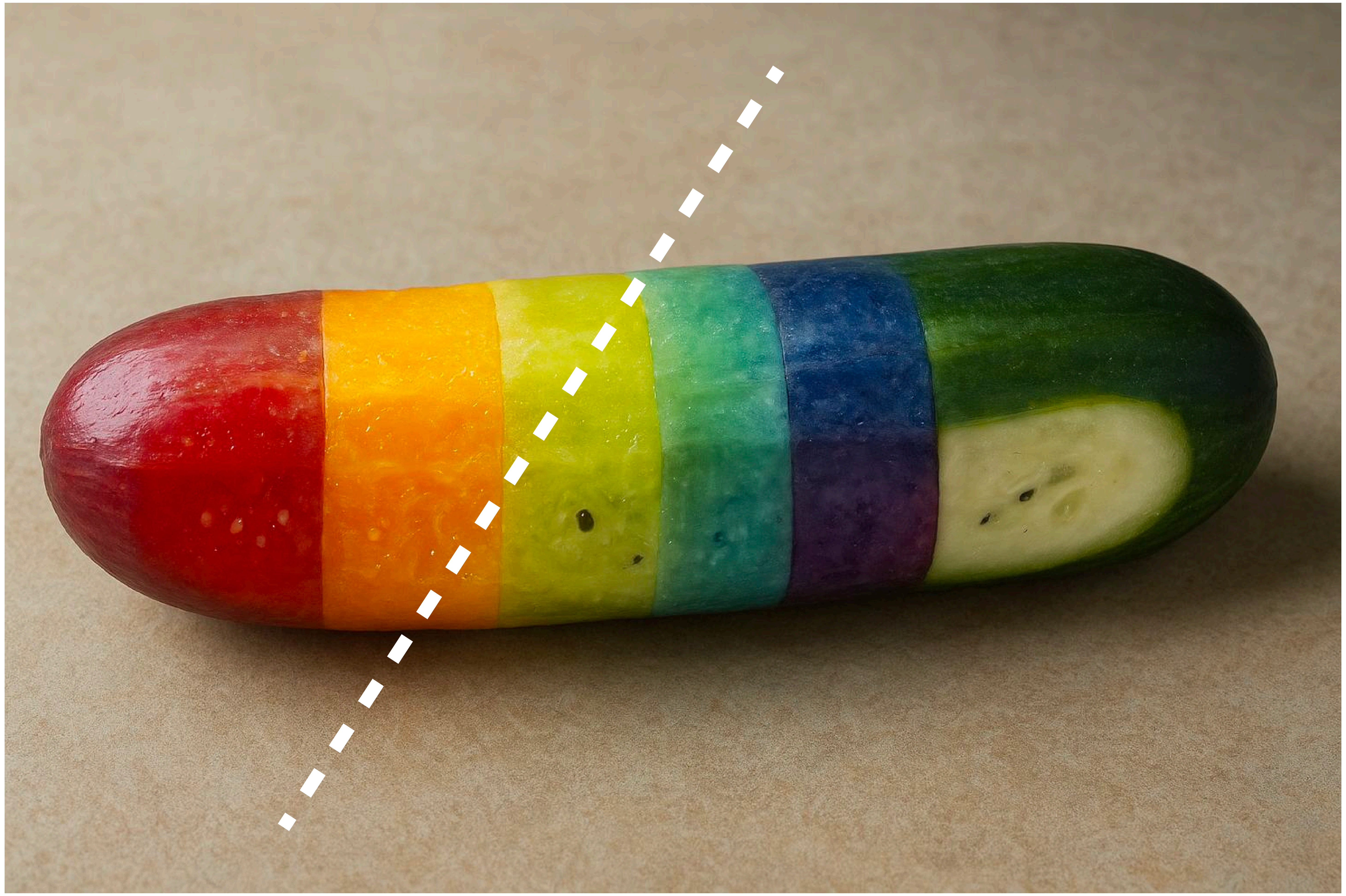

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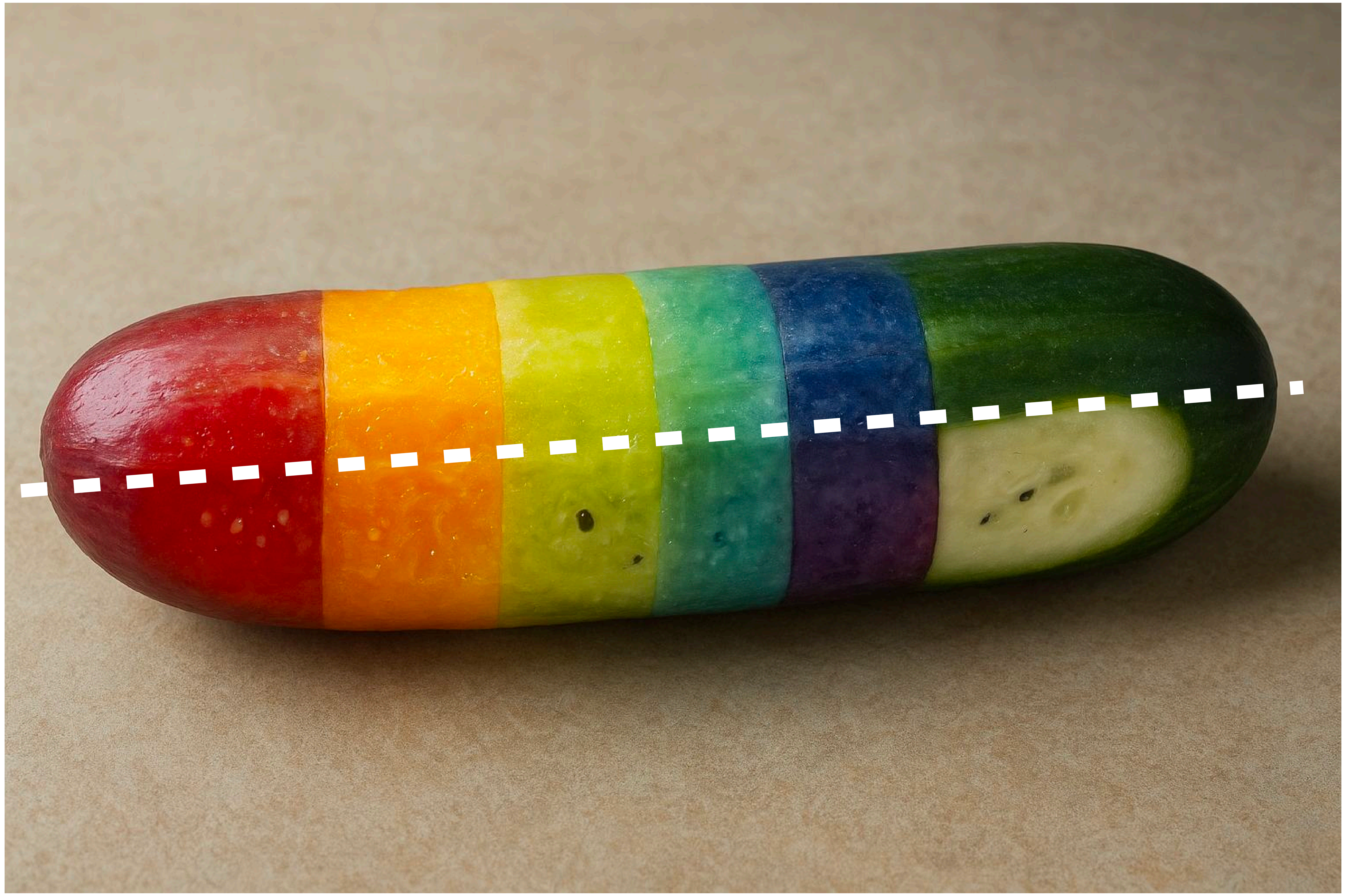

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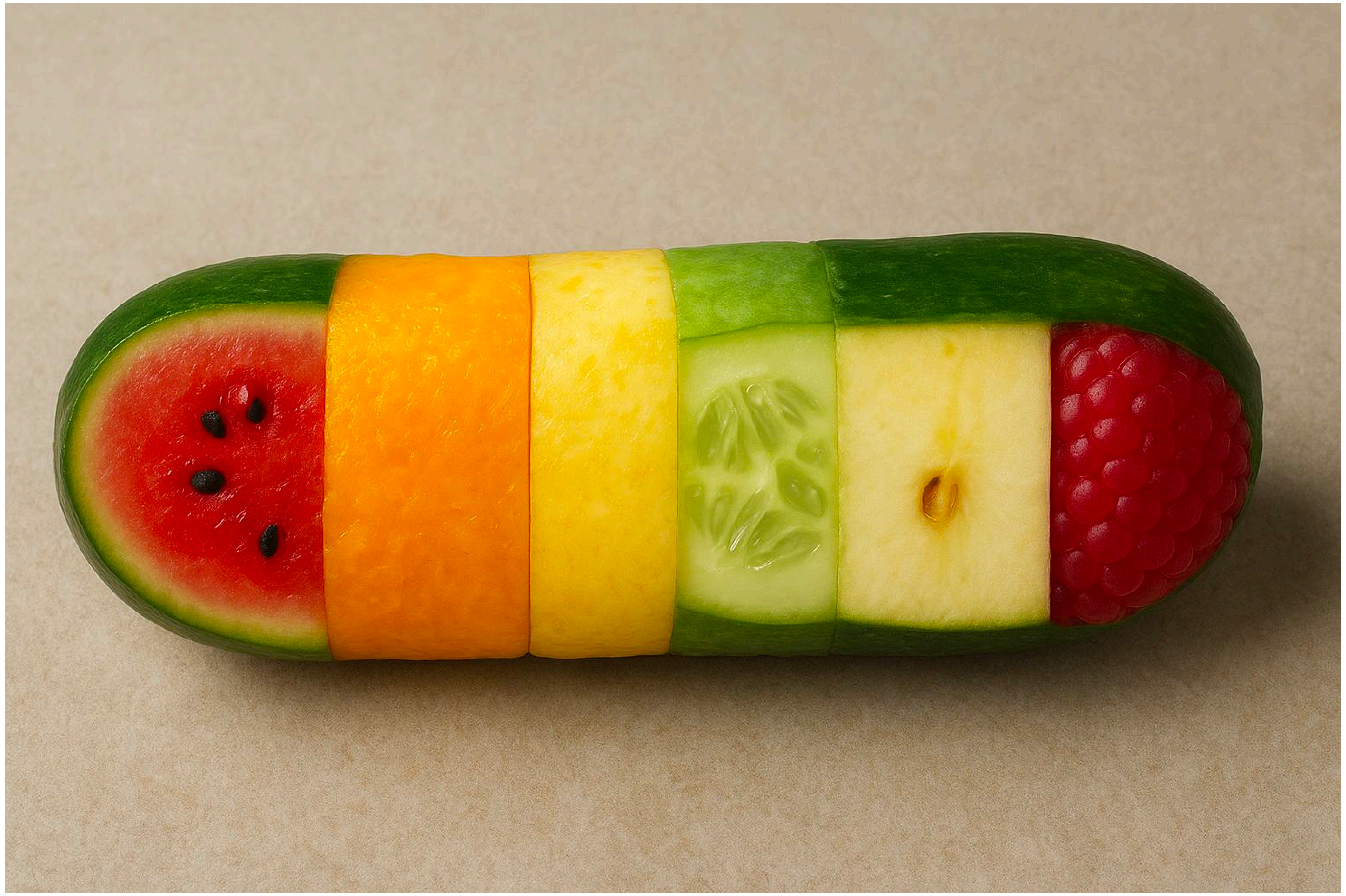
Step 2 - Dimensionality Reduction



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Step 2 - Dimensionality Reduction

Single-cell RNA sequencing (scRNA-seq) gives us a huge amount of information—measuring thousands of genes for thousands of cells.

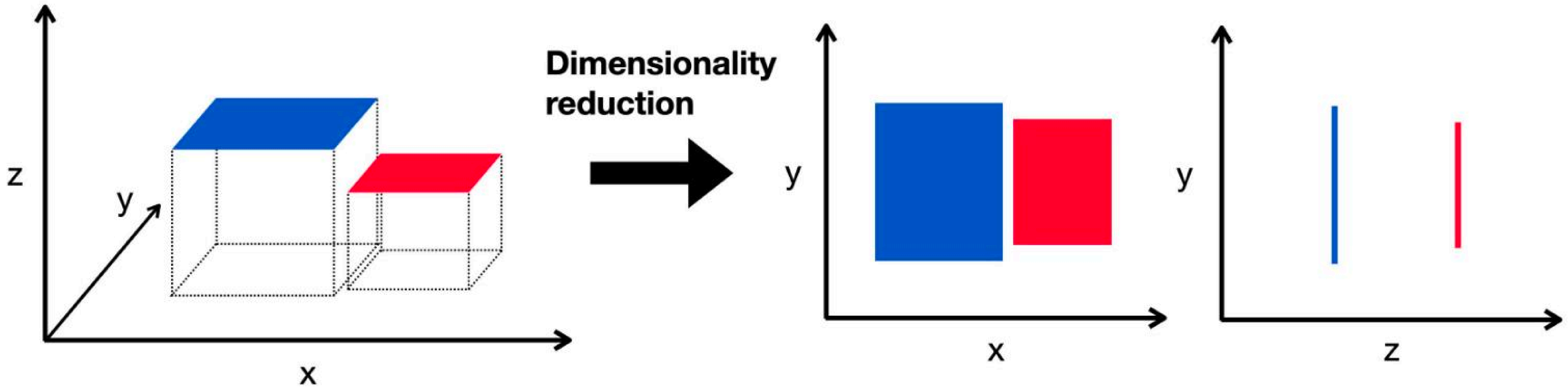
This data is high-dimensional, which makes it:

- Noisy: Full of irrelevant information.
- Complex: Difficult to see patterns.
- Hard to visualize: How do you plot 20,000 dimensions?

This is often called the "Curse of Dimensionality."

Step 2.1 - Dimensionality Reduction

The Goal: To simplify the data without losing the important biological information. We reduce thousands of dimensions (genes) to just a few (2 or 3) for visualization and analysis.



Step 2.1 - Dimensionality Reduction

Key Methods:

- PCA (Principal Component Analysis): A foundational method to capture the largest sources of variation in the data.
- t-SNE & UMAP: Powerful algorithms that are excellent for visualizing the data and revealing the underlying structure of cell populations.

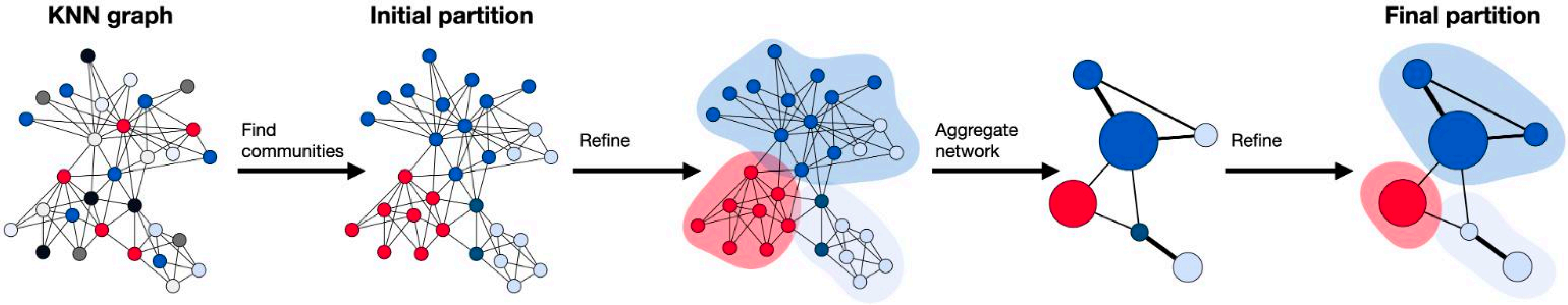
Think of it as creating a simple, accurate map from a massive, detailed globe.

will add a real plot from out data

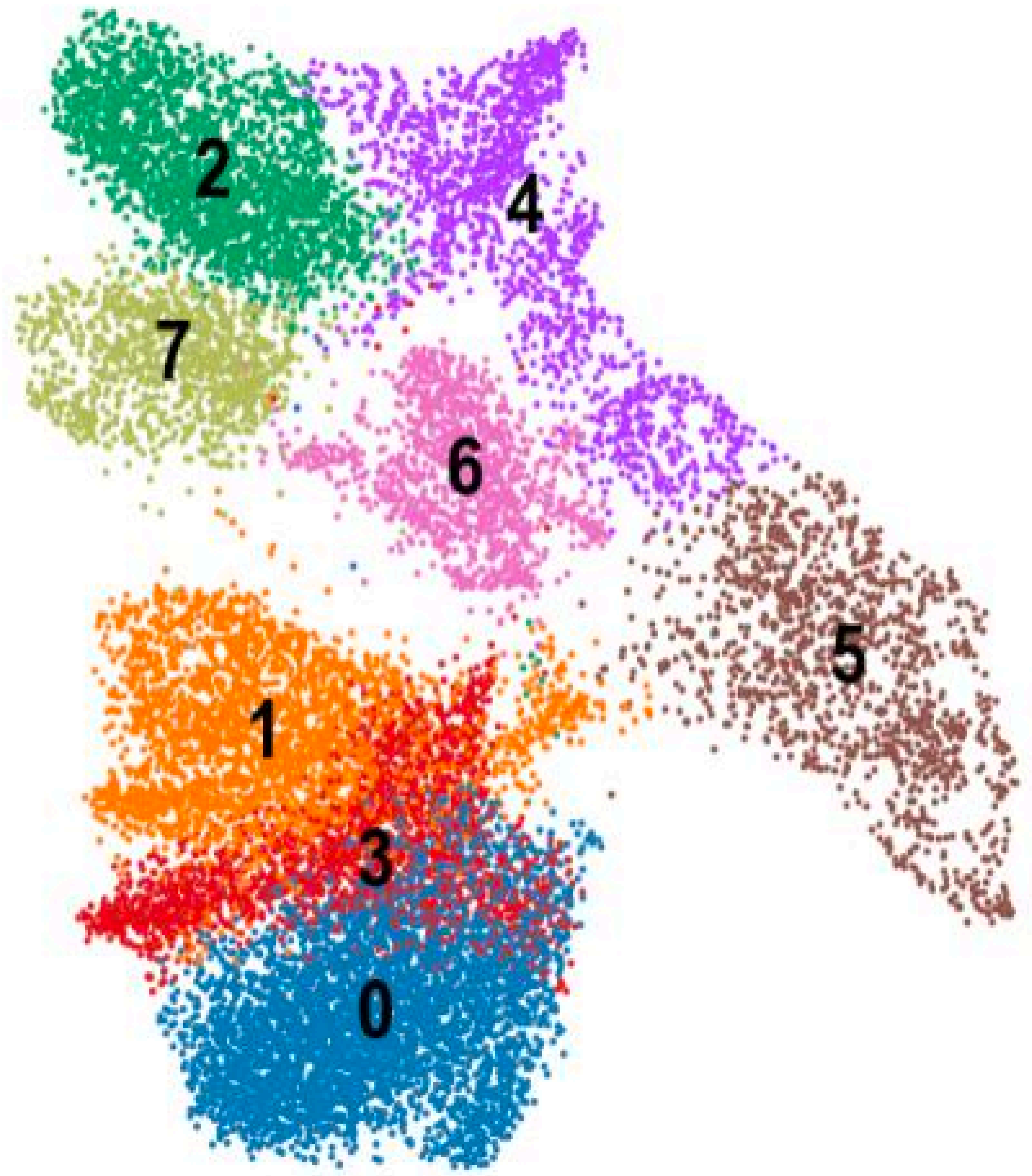
Step 2.2 - Clustering

The Goal Now that the data is simplified, we can group similar cells together. This is the first step in identifying different cell populations.

- Clustering algorithms look at the "map" we created (our UMAP or t-SNE plot).
- They group cells that are close to each other into distinct clusters.
- Each cluster represents a potential cell type or state.



At this stage, we have groups, but we don't know what they are yet.



Step 2.3 - How to annotate

How to Annotate? Two Main Approaches

Manual Annotation

1. Relies on expert knowledge of marker genes—genes known to be highly expressed in specific cell types.
2. Transparent and considered the "gold standard."
3. Can be subjective and time-consuming.

will add a real plot from out data

Step 2.3 - How to annotate

How to Annotate? Two Main Approaches

Automated Annotation

1. Uses computational algorithms and reference databases to automatically assign labels.
2. Pros: Fast, objective, and reproducible.
3. Only as good as the reference data; can be a "black box."

will add a real plot from out data

Acknowledgment

Special Thanks to my friend and lab mate Ariel

Also MICM team, especially Adrian

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